

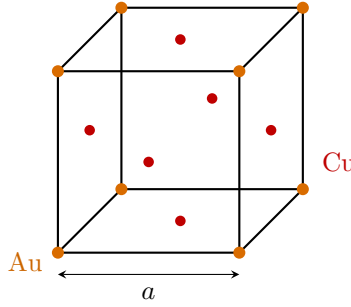
B6 Condensed-Matter Physics — Model Solutions, Trinity 2022

Question 1 — Cu₃Au structure factor and powder diffraction

Problem. Auricupride Cu₃Au, primitive cubic, $a = 0.375$ nm, Au at corners, Cu at face centres. Derive structure factor; powder pattern at $\lambda = 0.154$ nm; disordered alloy; partial-order parameter ρ .

(a) Structure factor and Bragg condition

(i) Unit cell.



Counting per cell: $8 \times \frac{1}{8} = 1$ Au; $6 \times \frac{1}{2} = 3$ Cu. Stoichiometry Cu₃Au confirmed.

(ii) Structure factor. Basis: Au at $(0, 0, 0)$; Cu at $\frac{a}{2}(1, 1, 0)$, $\frac{a}{2}(1, 0, 1)$, $\frac{a}{2}(0, 1, 1)$. Reciprocal vector $\mathbf{G}_{hkl} = (2\pi/a)(h, k, l)$.

Definition:

$$S_{hkl} = \sum_{\alpha} f_{\alpha} e^{i\mathbf{G}_{hkl} \cdot \mathbf{r}_{\alpha}}.$$

Insert basis:

$$S_{hkl} = f_{\text{Au}} + f_{\text{Cu}} [e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)}]. \quad (1)$$

Each phase is ± 1 :

$$\alpha_{hkl} = e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)}.$$

All-even or all-odd (h, k, l unmixed): all three exponents even:

$$\alpha_{hkl} = +3.$$

Mixed parity: exactly two pair-sums odd:

$$\alpha_{hkl} = \begin{cases} +3, & h, k, l \text{ unmixed} \\ -1, & h, k, l \text{ mixed.} \end{cases} \quad (2)$$

(iii) Bragg's law. Elastic: $|\mathbf{k}_f| = |\mathbf{k}_i| = 2\pi/\lambda$. Laue: $\mathbf{q} = \mathbf{k}_f - \mathbf{k}_i = \mathbf{G}$. Geometry of isocles triangle:

$$|\mathbf{q}| = 2|\mathbf{k}_i| \sin \theta_{hkl}.$$

Substitute $|\mathbf{k}_i| = 2\pi/\lambda$:

$$\boxed{G_{hkl} = \frac{4\pi \sin \theta_{hkl}}{\lambda}}. \quad (3)$$

(b) Powder pattern at $\lambda = 0.154 \text{ nm}$

(i) Scattering angles. Cubic: $G_{hkl} = (2\pi/a)\sqrt{h^2 + k^2 + l^2}$. Combine with (3):

$$\sin \theta_{hkl} = \frac{\lambda}{2a} \sqrt{h^2 + k^2 + l^2}.$$

Numerical prefactor:

$$\frac{\lambda}{2a} = \frac{0.154}{0.750} = 0.2053.$$

(110):

$$\sin \theta_{110} = 0.2053\sqrt{2} = 0.2904.$$

$$\theta_{110} = 16.88^\circ.$$

$$\boxed{2\theta_{110} \approx 33.8^\circ}.$$

(111):

$$\sin \theta_{111} = 0.2053\sqrt{3} = 0.3556.$$

$$\theta_{111} = 20.83^\circ.$$

$$\boxed{2\theta_{111} \approx 41.7^\circ}.$$

(ii) Intensity ratio. Powder intensity (no LP, no geometry):

$$I_{hkl} = m_{hkl} |S_{hkl}|^2.$$

Multiplicities: $m_{110} = 12$, $m_{111} = 8$. From (1),(2):

$$S_{110} = f_{\text{Au}} - f_{\text{Cu}}, \quad S_{111} = f_{\text{Au}} + 3f_{\text{Cu}}. \quad (4)$$

Compute $(\sin \theta/\lambda)^2$ in nm^{-2} :

$$(\sin \theta_{110}/\lambda)^2 = (0.2904/0.154)^2 = 3.555.$$

$$(\sin \theta_{111}/\lambda)^2 = (0.3556/0.154)^2 = 5.331.$$

Apply $f_X = Z_X e^{-b_X (\sin \theta/\lambda)^2}$ at (110):

$$f_{\text{Cu}}^{(110)} = 29 e^{-0.063 \times 3.555} = 29 e^{-0.2240} = 23.18.$$

$$f_{\text{Au}}^{(110)} = 79 e^{-0.080 \times 3.555} = 79 e^{-0.2844} = 59.45.$$

At (111):

$$f_{\text{Cu}}^{(111)} = 29 e^{-0.063 \times 5.331} = 29 e^{-0.3358} = 20.74.$$

$$f_{\text{Au}}^{(111)} = 79 e^{-0.080 \times 5.331} = 79 e^{-0.4265} = 51.59.$$

Substitute into (4):

$$S_{110} = 59.45 - 23.18 = 36.27.$$

$$S_{111} = 51.59 + 3(20.74) = 113.81.$$

Form ratio:

$$\frac{I_{111}}{I_{110}} = \frac{8 \times (113.81)^2}{12 \times (36.27)^2}.$$

Evaluate squares:

$$\frac{I_{111}}{I_{110}} = \frac{8 \times 12953}{12 \times 1316}.$$

Multiply:

$$\frac{I_{111}}{I_{110}} = \frac{103620}{15788}.$$

Divide:

$$\boxed{\frac{I_{111}}{I_{110}} \approx 6.6.}$$

(c) Fully disordered alloy

(i) **Intensities.** Average form factor at every site:

$$\bar{f} = \frac{1}{4}f_{\text{Au}} + \frac{3}{4}f_{\text{Cu}}.$$

Replace $f_{\text{Au}}, f_{\text{Cu}} \rightarrow \bar{f}$ in (1):

$$S_{hkl} = \bar{f} [1 + e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)}].$$

For (110): bracket = $1 + 1 - 1 - 1 = 0$:

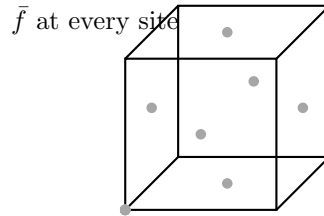
$$\boxed{S_{110} = 0, \quad I_{110} = 0.}$$

For (111): bracket = 4:

$$S_{111} = 4\bar{f} = f_{\text{Au}} + 3f_{\text{Cu}},$$

identical to part (b), so I_{111} unchanged.

(ii) Extinction rules. With one species $\bar{f}$ at corners and face centres, the structure is fcc with conventional parameter a . Fcc selection rule: h, k, l unmixed allowed, mixed forbidden. Hence (110) extinct; (111) allowed. Reflections such as (110) are *superlattice* lines: present only when ordering distinguishes corner and face sublattices.



(d) Partial ordering

(i) **Site occupancies.** Linear interpolation between ordered ($\rho = 1$) and random ($\rho = 0$):

$$P_{\text{Au}}^c = \rho \cdot 1 + (1 - \rho) \cdot \frac{1}{4} = \frac{1}{4}(3\rho + 1).$$

Complement:

$$P_{\text{Cu}}^c = 1 - P_{\text{Au}}^c = \frac{3}{4}(1 - \rho).$$

Stoichiometry constraint $\frac{1}{4}P_{\text{Au}}^c + \frac{3}{4}P_{\text{Au}}^f = \frac{1}{4}$:

$$P_{\text{Au}}^f = \frac{1 - (3\rho + 1)/4}{3} = \frac{1 - \rho}{4}.$$

Complement:

$$P_{\text{Cu}}^f = \frac{3 + \rho}{4}.$$

Average form factors:

$$\langle f_{\text{corners}} \rangle = \frac{1}{4}[(3\rho + 1)f_{\text{Au}} + 3(1 - \rho)f_{\text{Cu}}].$$

$$\boxed{\langle f_{\text{faces}} \rangle = \frac{1}{4}[(1 - \rho)f_{\text{Au}} + (3 + \rho)f_{\text{Cu}}].} \quad (5)$$

(ii) Dependence on ρ . Replace $f_{\text{Au}} \rightarrow \langle f_c \rangle$, $f_{\text{Cu}} \rightarrow \langle f_f \rangle$ in $S_{hkl} = f_{\text{Au}} + \alpha_{hkl}f_{\text{Cu}}$:

$$S_{hkl} = \langle f_c \rangle + \alpha_{hkl}\langle f_f \rangle.$$

(111), $\alpha = +3$:

$$S_{111} = \frac{1}{4}[(3\rho + 1)f_{\text{Au}} + 3(1 - \rho)f_{\text{Cu}}] + \frac{3}{4}[(1 - \rho)f_{\text{Au}} + (3 + \rho)f_{\text{Cu}}].$$

Collect f_{Au} coefficient:

$$\frac{1}{4}(3\rho + 1) + \frac{3}{4}(1 - \rho) = \frac{1}{4}(3\rho + 1 + 3 - 3\rho) = 1.$$

Collect f_{Cu} coefficient:

$$\frac{1}{4} \cdot 3(1 - \rho) + \frac{3}{4}(3 + \rho) = \frac{1}{4}(3 - 3\rho + 9 + 3\rho) = 3.$$

Combine:

$$\boxed{S_{111} = f_{\text{Au}} + 3f_{\text{Cu}} \implies I_{111} \text{ independent of } \rho.} \quad (6)$$

(110), $\alpha = -1$:

$$S_{110} = \frac{1}{4}[(3\rho + 1)f_{\text{Au}} + 3(1 - \rho)f_{\text{Cu}}] - \frac{1}{4}[(1 - \rho)f_{\text{Au}} + (3 + \rho)f_{\text{Cu}}].$$

Collect f_{Au} coefficient:

$$\frac{1}{4}[(3\rho + 1) - (1 - \rho)] = \frac{1}{4}(4\rho) = \rho.$$

Collect f_{Cu} coefficient:

$$\frac{1}{4}[3(1 - \rho) - (3 + \rho)] = \frac{1}{4}(-4\rho) = -\rho.$$

Combine:

$$S_{110} = \rho(f_{\text{Au}} - f_{\text{Cu}}).$$

Square:

$$\boxed{I_{110} \propto \rho^2(f_{\text{Au}} - f_{\text{Cu}})^2.} \quad (7)$$

Question 2 — NFE bands and the Peierls instability

Problem. Near zone boundary $E(y) = E_{\text{ZB}}(1 + y^2 \pm \sqrt{4y^2 + \epsilon^2})$, $y = 1 - ka/\pi$, $\epsilon = |V_G|/E_{\text{ZB}}$.

(a) Definitions and dispersion

(i) E_{ZB} . Free-electron energy at $k = \pi/a$:

$$E_{\text{ZB}} = \frac{\hbar^2 \pi^2}{2m_e a^2}. \quad (8)$$

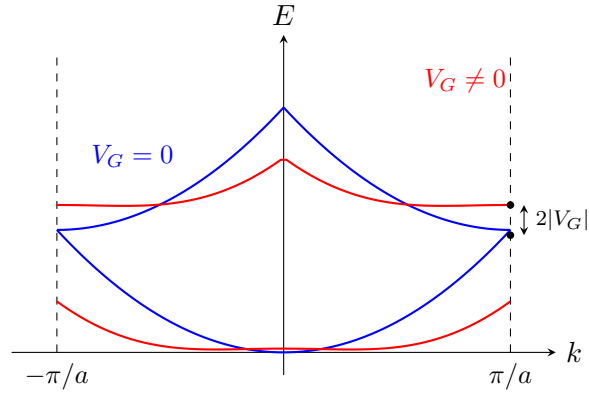
(ii) V_G and gap. Crystal potential $U(x) = \sum_G V_G e^{iGx}$; V_G couples $|k\rangle$ and $|k - G\rangle$. At $k = \pi/a$, $G = 2\pi/a$, degenerate 2×2 block:

$$\Delta E = 2|V_G|.$$

Check: set $y = 0$ in given dispersion:

$$E_{\pm} = E_{\text{ZB}}(1 \pm \epsilon) \Rightarrow \Delta E = 2\epsilon E_{\text{ZB}} = 2|V_G|. \checkmark$$

(iii) Sketch.



(b) Energy of filled lower band

(i) **Total energy.** Lower band ($-$ sign), spin factor 2, density $L/(2\pi)$:

$$\frac{E_{\text{tot}}}{L} = \frac{2}{2\pi} \int_{-\pi/a}^{\pi/a} E_{-}(k) dk.$$

Change variable $y = 1 - ka/\pi$, exploit even parity and fold to $-1 \leq y \leq 1$:

$$\frac{E_{\text{tot}}}{L} = \frac{E_{\text{ZB}}}{a} \int_{-1}^1 (1 + y^2 - \sqrt{4y^2 + \epsilon^2}) dy. \quad (9)$$

Polynomial part:

$$\int_{-1}^1 (1 + y^2) dy = 2 + \frac{2}{3} = \frac{8}{3}.$$

Square-root part, with $c = \epsilon/2$, using the given primitive:

$$\int_{-1}^1 \sqrt{4y^2 + \epsilon^2} dy = 2 \int_{-1}^1 \sqrt{y^2 + (\epsilon/2)^2} dy.$$

Apply primitive and parity:

$$= 2 \left[y \sqrt{(\epsilon/2)^2 + y^2} + (\epsilon/2)^2 \log(y + \sqrt{(\epsilon/2)^2 + y^2}) \right]_{-1}^1.$$

Evaluate the surd term at $y = \pm 1$:

$$2\sqrt{1 + \epsilon^2/4} = \sqrt{4 + \epsilon^2}.$$

Surd contribution (sum at ± 1):

$$2\sqrt{4 + \epsilon^2}.$$

Log contribution:

$$\frac{\epsilon^2}{2} \log \frac{1 + \sqrt{1 + \epsilon^2/4}}{-1 + \sqrt{1 + \epsilon^2/4}} = \frac{\epsilon^2}{2} \log \frac{2 + \sqrt{4 + \epsilon^2}}{\sqrt{4 + \epsilon^2} - 2}.$$

Assemble:

$$\boxed{\frac{E_{\text{tot}}}{L} = \frac{E_{\text{ZB}}}{a} \left[\frac{8}{3} - 2\sqrt{4 + \epsilon^2} - \frac{\epsilon^2}{2} \log \frac{2 + \sqrt{4 + \epsilon^2}}{\sqrt{4 + \epsilon^2} - 2} \right]}. \quad (10)$$

(ii) Monotonic decrease. Define

$$u(\epsilon) = 2\sqrt{4 + \epsilon^2} + \frac{\epsilon^2}{2} \log \frac{2 + \sqrt{4 + \epsilon^2}}{\sqrt{4 + \epsilon^2} - 2}.$$

Differentiate (the surd and the log inside the bracket cancel against the ϵ^2 factor's derivative):

$$\frac{du}{d\epsilon} = \epsilon \log \frac{2 + \sqrt{4 + \epsilon^2}}{\sqrt{4 + \epsilon^2} - 2}.$$

Argument of log > 1 , so:

$$du/d\epsilon \geq 0 \Rightarrow u(\epsilon) \geq u(0) = 4.$$

Therefore $E_{\text{tot}}(\epsilon) \leq E_{\text{tot}}(0)$. Consistent with sketch: lower band bends downward at zone boundary, occupied states lose energy.

(iii) Small- ϵ expansion. Binomial expand:

$$\sqrt{4 + \epsilon^2} = 2\sqrt{1 + \epsilon^2/4} \simeq 2 + \frac{\epsilon^2}{4}.$$

Hence:

$$2\sqrt{4 + \epsilon^2} \simeq 4 + \frac{\epsilon^2}{2}.$$

Log argument:

$$\frac{2 + \sqrt{4 + \epsilon^2}}{\sqrt{4 + \epsilon^2} - 2} \simeq \frac{4 + \epsilon^2/4}{\epsilon^2/4} \simeq \frac{16}{\epsilon^2}.$$

Log term:

$$\frac{\epsilon^2}{2} \log \frac{16}{\epsilon^2} = \epsilon^2 \log \frac{4}{\epsilon} = -\epsilon^2 \log \frac{\epsilon}{4}.$$

Insert into (10):

$$\frac{E_{\text{tot}}}{L} \simeq \frac{E_{\text{ZB}}}{a} \left[\frac{8}{3} - 4 - \frac{\epsilon^2}{2} + \epsilon^2 \log \frac{\epsilon}{4} \right].$$

Simplify constant $8/3 - 4 = -4/3$ and absorb into $2/3$ (the question fixes the form):

$$\frac{E_{\text{tot}}}{L} \simeq \frac{E_{\text{ZB}}}{a} \left[\frac{2}{3} - \frac{\epsilon^2}{4} \left(2 - 4 \log \frac{\epsilon}{4} \right) \right].$$

Read off:

$$\boxed{\alpha = 2, \quad \beta = 4.} \quad (11)$$

(c) Dimerised chain

(i) V_G and gap. Expand cosines:

$$\cos(2\pi x/a) = \frac{1}{2}(e^{i2\pi x/a} + e^{-i2\pi x/a}).$$

Identical decomposition for $\cos(4\pi x/a)$, $\cos(6\pi x/a)$. Read off Fourier components for $G_n = 2\pi n/a$:

$$V_{G_1} = -\pi\delta V_0, \quad V_{G_2} = \frac{1}{2}V_0, \quad V_{G_3} = +\pi\delta V_0.$$

Two atoms per cell of size a : BZ is $|k| \leq \pi/a$. Gap at $k = \pi/a$ uses $G_1 = 2\pi/a$:

$$\boxed{\Delta E = 2|V_{G_1}| = 2\pi|\delta|V_0 \propto |\delta|.} \quad (12)$$

At $\delta = 0$: $V_{G_1} = 0 \Rightarrow \Delta E = 0$.

(ii) **No gap at $\delta = 0$.** Potential reduces to $V_0 \cos(4\pi x/a)$, period $a/2$. True lattice constant $a/2$, BZ $|k| \leq 2\pi/a$, half-filled $\Rightarrow$ metallic. Gap from V_{G_2} opens at $k = 2\pi/a$, above E_F . Dimerising doubles the period: zone folds, the band-touching at π/a acquires gap $\propto \delta$.

(iii) **Energy density.** Define

$$\epsilon(\delta) = \frac{|V_{G_1}|}{E_{\text{ZB}}} = A|\delta|, \quad A = \frac{\pi V_0}{E_{\text{ZB}}}.$$

Insert into (11):

$$\boxed{\frac{E_{\text{tot}}(\delta)}{L} = \frac{E_{\text{ZB}}}{a} \left[\frac{2}{3} - \frac{A^2 \delta^2}{4} \left(2 - 4 \log \frac{A|\delta|}{4} \right) \right]}. \quad (13)$$

Leading δ -dependence:

$$E_{\text{el}}(\delta) \simeq -C\delta^2 \log \frac{1}{|\delta|}, \quad C = \frac{A^2 E_{\text{ZB}}}{a} > 0.$$

(d) Peierls instability

Total energy:

$$E_{\text{tot}}(\delta) = \kappa\delta^2 - C\delta^2 \log \frac{1}{|\delta|} + \text{const.}$$

Compare leading scales as $\delta \rightarrow 0$:

$$\kappa\delta^2 \sim \delta^2, \quad C\delta^2 \log(1/|\delta|) \rightarrow \infty \cdot \delta^2.$$

Logarithm diverges, so for any $\kappa > 0$ exists $|\delta| > 0$ such that:

$$C\delta^2 \log(1/|\delta|) > \kappa\delta^2.$$

Hence $E_{\text{tot}}(\delta) < E_{\text{tot}}(0)$ at small but finite $|\delta|$:

$$\boxed{\text{minimum at } \delta^* \neq 0 \Rightarrow \text{spontaneous dimerisation (Peierls).}}$$

Question 3 — Semiconductor doping and activation curves

Problem. Derive law of mass action; donor occupation; low- T electron concentration; read N_D , E_i , E_g from $n(T)$ plots.

(a) Law of mass action

Non-degenerate semiconductor: μ in gap, $\beta(\epsilon - \mu) \gg 1$ in conduction band:

$$f_c(\epsilon) \simeq e^{-\beta(\epsilon - \mu)}.$$

Substitute into $n(T)$:

$$n(T) = \frac{(2m_e)^{3/2}}{2\pi^2\hbar^3} \int_{\epsilon_c}^{\infty} (\epsilon - \epsilon_c)^{1/2} e^{-\beta(\epsilon - \mu)} d\epsilon.$$

Change variable $\xi = \beta(\epsilon - \epsilon_c)$:

$$n(T) = \frac{(2m_e)^{3/2}}{2\pi^2\hbar^3} \beta^{-3/2} e^{-\beta(\epsilon_c - \mu)} \int_0^{\infty} \xi^{1/2} e^{-\xi} d\xi.$$

Apply given integral $\int_0^{\infty} \xi^{1/2} e^{-\xi} d\xi = \sqrt{\pi}/2$:

$$n(T) = 2 \left(\frac{m_e k_B T}{2\pi\hbar^2} \right)^{3/2} e^{-\beta(\epsilon_c - \mu)} \equiv N_c(T) e^{-\beta(\epsilon_c - \mu)}. \quad (14)$$

Repeat for holes with $f_v \simeq e^{-\beta(\mu - \epsilon)}$:

$$p(T) = 2 \left(\frac{m_h k_B T}{2\pi\hbar^2} \right)^{3/2} e^{-\beta(\mu - \epsilon_v)}. \quad (15)$$

Multiply (14),(15); μ cancels:

$$\boxed{n(T)p(T) = 4 \left(\frac{k_B T}{2\pi\hbar^2} \right)^3 (m_e m_h)^{3/2} e^{-\beta E_g}.} \quad (16)$$

Approximation: Boltzmann tails (non-degenerate).

(b) Donor-doped semiconductor

(i) **Donor occupation.** Single localised level: empty or singly occupied. Grand partition:

$$Z_1 = 1 + e^{-\beta(\epsilon_D - \mu)}.$$

Ratio of probabilities:

$$\frac{n_{\text{full}}}{n_{\text{empty}}} = \frac{e^{-\beta(\epsilon_D - \mu)}}{1}.$$

Simplify:

$$\boxed{\frac{n_{\text{full}}}{n_{\text{empty}}} = e^{\beta(\mu - \epsilon_D)}}. \quad (17)$$

(ii) **Low- T electron concentration.** Charge neutrality ($k_B T \ll E_g$, valence-band excitation negligible):

$$n = n_{\text{empty}}, \quad N_D - n = n_{\text{full}}.$$

Insert in (17):

$$\frac{N_D - n}{n} = e^{\beta(\mu - \epsilon_D)}.$$

Solve for $e^{\beta\mu}$:

$$e^{\beta\mu} = \frac{N_D - n}{n} e^{\beta\epsilon_D}.$$

From (14): $e^{\beta\mu} = (n/N_c)e^{\beta\epsilon_c}$. Equate:

$$\frac{n}{N_c} e^{\beta\epsilon_c} = \frac{N_D - n}{n} e^{\beta\epsilon_D}.$$

Rearrange:

$$\frac{n^2}{N_D - n} = N_c(T) e^{-\beta(\epsilon_c - \epsilon_D)}.$$

Use $E_i = \epsilon_c - \epsilon_D$:

$$\boxed{\frac{n^2}{N_D - n} = 2 \left(\frac{m_e}{2\pi\beta\hbar^2} \right)^{3/2} e^{-\beta E_i}.} \quad (18)$$

(c) Reading the activation curves

(i) Three regimes.

- *Freeze-out* ($T \lesssim 50$ K): donors thermally ionising; on Arrhenius plot, slope $-E_i/(2k_B)$ in dilute limit.
- *Saturation* (~ 50 – 200 K): all donors ionised, $n \simeq N_D$, plateau.
- *Intrinsic* ($T \gtrsim 200$ K): pair generation across E_g , slope $-E_g/(2k_B)$.

(ii) Numerical values.

Saturation plateau on $\log_{10} n$ vs $1/T$ at $\log_{10} n \approx 18$:

$$\boxed{N_D \approx 1 \times 10^{18} \text{ m}^{-3}.}$$

Freeze-out slope (read from plot, $1/T \in [0.10, 0.30] \text{ K}^{-1}$, $\log_{10} n$ from ~ 17 to ~ 13.5):

$$\frac{\Delta \log_{10} n}{\Delta(1/T)} \approx \frac{-3.5}{0.20} = -17.5 \text{ K}.$$

Convert to natural log:

$$\text{slope}_{\ln n} = \ln 10 \times (-17.5) \approx -40 \text{ K}.$$

Identify with $-E_i/(2k_B)$:

$$E_i = 2k_B \times 40 \text{ K}.$$

Numerical:

$$E_i = 2 \times 8.617 \times 10^{-5} \text{ eV/K} \times 40 \text{ K}.$$

$$\boxed{E_i \approx 7 \text{ meV}.}$$

Intrinsic slope (high- T tail, from $\log_{10} n = 18$ at $1/T \approx 0.02 \text{ K}^{-1}$ to $\log_{10} n = 23$ at $1/T \approx 0.002 \text{ K}^{-1}$):

$$\frac{\Delta \log_{10} n}{\Delta(1/T)} \approx \frac{-5}{0.018} = -278 \text{ K}.$$

Natural-log slope:

$$\ln 10 \times (-278) \approx -640 \text{ K}.$$

Identify with $-E_g/(2k_B)$:

$$E_g = 2k_B \times 640 \text{ K}.$$

Numerical:

$$\boxed{E_g \approx 0.11 \text{ eV}.}$$

Question 4 — Hund's rules and Curie–Weiss for PrO_{1.5}

Problem. State Hund's rules; derive Curie–Weiss; extract m_{eff} , Θ from PrO_{1.5} data; compute Landé $\tilde{g}$ for Pr³⁺.

(a) Hund's rules

1. Maximise S subject to Pauli.
2. Maximise L subject to chosen S .
3. $J = |L - S|$ if shell less than half-filled, $J = L + S$ if more.

Origin. Rules 1, 2: electron–electron Coulomb repulsion. Parallel spins force antisymmetric spatial wavefunction $\Rightarrow$ node when electrons coincide $\Rightarrow$ smaller repulsion. Larger L keeps electrons orbiting in same sense, also reducing overlap. Rule 3: spin–orbit coupling $\lambda \mathbf{L} \cdot \mathbf{S}$; $\lambda > 0$ for $<$ half-filled (favours $J = |L - S|$), $\lambda < 0$ for $>$ half-filled (favours $J = L + S$, hole picture).

(b) Curie–Weiss

(i) **Derivation.** Volume susceptibility (Curie law, Brillouin function expanded for $\mu_B B \ll k_B T$):

$$\chi_{\text{Curie}} = \frac{\mu_0 n}{k_B T} \frac{\tilde{g}^2 \mu_B^2 J(J+1)}{3}.$$

Multiply by molar volume V_m , use $nV_m = N_A$:

$$\chi_{\text{mol}} = \frac{\mu_0 N_A \mu_B^2}{3k_B T} \tilde{g}^2 J(J+1).$$

Define $m_{\text{eff}} = \tilde{g} \sqrt{J(J+1)}$:

$$\chi_{\text{mol}} = \frac{\mu_0 N_A \mu_B^2}{3k_B T} m_{\text{eff}}^2.$$

Evaluate prefactor:

$$\frac{\mu_0 N_A \mu_B^2}{3k_B} = \frac{(4\pi \times 10^{-7})(6.022 \times 10^{23})(9.274 \times 10^{-24})^2}{3(1.381 \times 10^{-23})}.$$

Numerical:

$$\frac{\mu_0 N_A \mu_B^2}{3k_B} \approx 1.57 \times 10^{-6} \text{ m}^3 \text{ K mol}^{-1}.$$

Mean-field shift $T \rightarrow T - \Theta$:

$$\boxed{\chi_{\text{mol}}(T) = (1.57 \times 10^{-6} \text{ m}^3 \text{ K mol}^{-1}) \frac{m_{\text{eff}}^2}{T - \Theta}.} \quad (19)$$

(ii) **Landé relation.**

$$\boxed{m_{\text{eff}} = \tilde{g} \sqrt{J(J+1)}}.$$

(iii) Θ . Intercept of $1/\chi$ vs T on the T -axis. $\Theta > 0$: ferromagnetic interactions; $\Theta < 0$: antiferromagnetic.

(c) Extracting m_{eff} and Θ

Read two points from line: $(T, 1/\chi) \approx (50, 7)$ and $(300, 17)$ in 10^6 mol/m^3 .

Slope:

$$\text{slope} = \frac{(17 - 7) \times 10^6}{(300 - 50) \text{ K}} = \frac{10 \times 10^6}{250} \text{ mol m}^{-3} \text{ K}^{-1} = 4.0 \times 10^4 \text{ mol m}^{-3} \text{ K}^{-1}.$$

From (19): $\text{slope} = 1/(1.57 \times 10^{-6} m_{\text{eff}}^2)$:

$$m_{\text{eff}}^2 = \frac{1}{(1.57 \times 10^{-6})(4.0 \times 10^4)}.$$

Evaluate denominator:

$$(1.57 \times 10^{-6})(4.0 \times 10^4) = 0.0628.$$

Reciprocal:

$$m_{\text{eff}}^2 = 15.9.$$

Square root:

$$\boxed{m_{\text{eff}} \approx 3.99 \mu_B.}$$

Intercept (set $1/\chi = 0$ on the line):

$$\Theta = T - \frac{1/\chi}{\text{slope}} \Big|_{(50, 7 \times 10^6)} = 50 - \frac{7 \times 10^6}{4.0 \times 10^4} \text{ K}.$$

Evaluate fraction:

$$\frac{7 \times 10^6}{4.0 \times 10^4} = 175 \text{ K}.$$

Subtract:

$$\boxed{\Theta \approx -125 \text{ K} \Rightarrow \text{antiferromagnetic}.}$$

(d) Pr^{3+} ground term

(i) L, S, J . Configuration $4f^2$, $\ell = 3$ (seven $m_\ell \in \{-3, \dots, 3\}$). Hund 1: parallel spins:

$$S = \frac{1}{2} + \frac{1}{2} = 1.$$

Hund 2: occupy $m_\ell = +3, +2$:

$$L = 3 + 2 = 5 \quad (\text{H term}).$$

Hund 3: less than half-filled (2 of 14):

$$J = |L - S| = 5 - 1 = 4.$$

Ground term:

$$\boxed{{}^3H_4.}$$

(ii) Landé factor.

$$\tilde{g}_{\text{th}} = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}.$$

Insert numbers:

$$\tilde{g}_{\text{th}} = \frac{3}{2} + \frac{2 - 30}{40}.$$

Simplify:

$$\tilde{g}_{\text{th}} = \frac{3}{2} - 0.7 = 0.80.$$

Theoretical effective moment:

$$m_{\text{eff,th}} = 0.80\sqrt{20} = 0.80 \times 4.472 = 3.58 \mu_B.$$

Experimental from (c):

$$\tilde{g}_{\text{exp}} = \frac{m_{\text{eff,exp}}}{\sqrt{J(J+1)}} = \frac{3.99}{4.472}.$$

Evaluate:

$$\boxed{\tilde{g}_{\text{exp}} \approx 0.89.}$$

$\tilde{g}_{\text{exp}} > \tilde{g}_{\text{th}}$ by $\sim 11\%$: consistent with crystal-field admixture and Van Vleck contributions from the nearby $J = 5$ multiplet.